Apartments

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1084 | Sorting and Searching | https://cses.fi/problemset/task/1084

There are n applicants and m free apartments. Your task is to distribute the apartments so that as many applicants as possible will get an apartment.

Each applicant has a desired apartment size, and they will accept any apartment whose size is close enough to the desired size.

Input

The first input line has three integers n, m, and k: the number of applicants, the number of apartments, and the maximum allowed difference.

The next line contains n integers $a_1, a_2, \ldots, a_n$: the desired apartment size of each applicant. If the desired size of an applicant is x, they will accept any apartment whose size is between $x-k$ and $x+k$.

The last line contains m integers $b_1, b_2, \ldots, b_m$: the size of each apartment.

Output

Print one integer: the number of applicants who will get an apartment.

Constraints

- $1 \le n, m \le 2 \cdot 10^5$
- $0 \le k \le 10^9$
- $1 \le a_i, b_i \le 10^9$

Example

Input

```
4 3 5
60 45 80 60
30 60 75
```

Output

```
2
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